California State University, Long Beach

Mechanical and Aerospace Engineering Department

MAE 300: Engineering Instrumentation and Measurement

Experiment No. 3

Temperature Measurement

Lab Section 19

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Abstract:

The objective of this experiment is for students to become familiar with different thermometers and their corresponding response times. Five different thermometers were used: alcohol-in-glass, platinum RTD, thermistor, thermocouple, and bi-metal. The alcohol-in-glass measurements were used as a reference for each of the other four thermometers. The corresponding calibration equations, standard errors, correlation coefficients, and time constants were calculated using the method of least squares and linear interpolation. Each of the thermometers measured the temperature of water being boiled, with a final temperature of 90 °C. A total of 90 data points were required for each of the data sets and the method of least squares were used to find the calibration calculations for the platinum RTD, thermistor, thermocouple, and bimetal. The thermal response times were found by finding the equivalent alcohol-in-glass temperature for each of the other four thermometers. The time constants for each of the thermometers correspond to the response times for each thermometer.

Introduction:

Five different types of thermometers were used in this experiment, each with distinct mechanical properties pertaining to the response time of the temperature measurements. The alcohol-in-glass thermometer measures temperature through thermal expansion. Alcohol has a relatively large thermal expansion coefficient, low freezing point, and low boiling point. This low boiling point limits its reliability in measuring boiling water. Another drawback comes from its nonlinear expansion, which can make readings inconsistent and less precise. Resistance temperature detectors (RTDs) and thermistors operate similarly by using the resistance of certain metals to measure temperature. Platinum RTDs provide the most stable relationship between resistance and temperature over a large temperature range compared to other substances. Compared to platinum RTDs, thermistors are less stable and have a smaller temperature range but are more sensitive to small temperature changes and have a faster response time. Thermocouples

Each of these different thermometers have different response times based on their mechanical properties. The response time of a thermometer is a very crucial value in determining the type of thermometer that should be used in different applications. The response time reflects how quickly the thermometer reaches thermal equilibrium of the substance of interest. A faster response time will directly correlate with accurate real-time readings. It is for this reason that learning how to calculate the time constant is crucial when dealing with apparatuses that measure temperature.

Background & Theory:

Temperature is defined as the thermal potential of a given object or substance. To investigate how different sensors detect and respond to temperature changes, this experiment makes use of five measurement devices: Thermometer that uses alcohol, a thermocouple, a thermistor, a platinum resistance thermometer (RTD), and a bimetal thermometer. For each of

the different thermometer temperature measurements, the calibration equations, a_0, a_1, correlation coefficient, and standard error are calculated using (1), (2), (3), and (5), respectively.

The mercury thermometer operates on the principal of thermal expansion, where liquid mercury increases in volume as temperature increases, allowing a direct visual reading along a calibrated scale. A thermocouple functions differently, generating a small electro-motive-force (EMF) at the junction of the two different metal wires when a temperature difference exists between the measurement point and a reference point. This voltage is then converted to the corresponding temperature value. A thermistor is a semi-conductor-based resistor that has a highly sensitive electrical resistance that changes significantly due to temperature. The thermistor operates best between a range of -100 to 250 degrees Celsius (°C). The platinum RTD leverages the stable and linear relationship between temperature and the resistance of pure platinum wire, which registers 100 ohms at 0°C and is encased in protective insulation and a metallic shell for durability. The bimetal thermometer consists of two bonded metal strips with different thermal expansion rates which causes the combined strip to bend mechanically when exposed to heat, with the degree of deflection indicating the temperature on a dial.

The thermal response from these devices are crucial to any given experiment since the reponse time isn't instantaneous, instead these thermoters follow a pretidtable expemential curve to calculate the new temperature. Therefore in order to analyze this behavior, the raw sensor must first be corrected using the calibration equation, where the corrected temperature is found by solving for x using the least squares fit parameters (7). Once the corrected values have been calculated, the thermal response is characterized by identifying the temperature that marks 63.2% of the total change from the intial temperature T_i to the final equilibrium temperature T_f using (8). This 63.2% threshold arises naturally from the fact that the model of the data is similar to the half-life exponential model. The value of 0.632 or 63.2% of the total change in temperature comes from the value of: $\ln(\frac{1}{2})$. Which corresponds precisely to the point where one time constant has elapsed. Since experimental data is collected at discrete intervals, the exact moment the sensor crosses this threshold is determined through linear interpolation between the two nearest recorded data points (9). The resulting value t_c is the time constant for the specific sensor. The smaller the value results in a faster thermal response, while a larger value indicates a sensor that is slower to equilibrate with its surrounding environment.

(1) $y = a_0 + a_i x$

(2) $a_0 = \frac{\Sigma x_i \Sigma x_i y_i - \Sigma y_i \Sigma x_i^2}{(\Sigma x_i)^2 - n(\Sigma x_i)^2}$

(3) $a_1 = \frac{\Sigma x_i \Sigma y_i - n\Sigma x_i y_i}{(\Sigma x_i)^2 - n\Sigma x_i^2}$

(4) $\varepsilon_i = (y_i - (a_0 + a_1 x_i))^2$

(5) $SE = \sqrt{\frac{\Sigma(y_i-(a_0+a_1x))^2}{n-2}}$

(6) $r = \frac{n(\Sigma x_i y_i)-(\Sigma x_i)(\Sigma y_i)}{\sqrt{(n\Sigma x_i{}^2)-(\Sigma x_i)^2)(n\Sigma y_i{}^2)-(\Sigma y_i)^2)}}$

(7) $x = \frac{(y-a_0)}{a_1}$

(8) $T_{63.2\%\,SC} = T_i + 0.632 * (T_f - T_i)$

(9) $t_c = t_1 + \frac{T_{63.2\%\,SC}-T_1}{T_2-T_1}(t_2 - t_1)$

Experimental Procedures:

To begin the experiment, the alcohol-in-glass, thermocouple, thermistor, platinum RTD, and the bi-metal thermometers were placed in pot containing room temperature water to calibrate the instruments. The heater was then turned on to warm up the water in the pot to boil. With the water heater, the temperature vs time was recorded on video on all thermometers for at least five steps from room temperature to boiling point temperature. The thermometers were placed through a plate, to reduce the amount of heat loss as illustrated in Figure 1.1. A total of 91 data points were extracted from the video recordings. Once all the data was compiled, the thermocouple, thermistor, platinum RTD, and the bi-metal thermometers for thermal responses were compared to the alcohol-in-glass. This was done to get the corrected temperatures with the calibration equation (7) for the metal thermometers, since metal has a higher thermal conductivity and glass casing provides thermal lag.

Figure 1.1: Alcohol-in-glass Thermometer and Bi-metal Thermometer Measurement Recording



Figure 1.2: Thermistor, Thermocouple, and Platinum RTD Measurement Recording



Raw data:

Table 1.1: Temperature Measurements

Alcohol in Glass	Platinum	Thermistor	Thermocouple	Bimetal
x (°C)	y_pt (°C)	y_thermistor (°C)	y_TC (°C)	y_Bimetal (°C)
20.00	12.50	11.70	20	20.00
20.00	12.90	13.60	20	20.00
20.00	13.70	15.10	20	20.00
20.00	15.20	16.20	19	20.00
20.00	17.00	16.60	18	20.00
20.00	18.30	17.00	20	20.00
20.00	19.40	17.30	23	20.00
22.00	19.90	18.10	22	20.00
22.00	20.90	18.70	22	20.00
24.00	21.40	20.10	24	20.00

24.00	21.80	21.90	23	20.00
28.00	22.50	22.20	23	20.00
30.00	23.10	22.80	26	20.00
30.00	23.70	24.10	24	20.00
30.00	24.40	24.80	26	20.00
30.00	25.00	25.20	27	20.00
30.00	25.60	26.40	28	20.00
30.00	26.30	26.40	28	20.00
30.00	26.90	26.50	29	20.00
30.00	27.70	27.50	30	20.00
33.00	28.70	28.30	31	20.00
35.00	29.70	29.20	31	20.00
35.00	30.20	29.40	31	20.00
38.00	31.40	30.50	33	20.00
40.00	32.00	31.00	34	20.00
40.00	34.20	31.70	32	22.00
42.00	34.90	32.80	33	25.00
42.00	35.60	33.40	37	25.00
42.00	36.70	34.90	37	26.00
42.00	38.00	36.40	37	26.00
42.00	38.80	36.70	38	28.00
45.00	39.40	37.30	39	30.00
45.00	40.50	38.50	43	33.00
45.00	41.10	38.70	44	33.00
45.00	42.40	39.90	42	35.00
47.00	43.90	41.10	46	35.00
47.00	44.20	41.60	47	35.00
47.00	44.80	42.60	45	35.00
47.00	45.40	43.20	45	39.00
47.00	46.30	44.50	47	40.00
48.00	47.00	45.10	47	45.00
50.00	47.80	45.20	48	47.00
50.00	48.60	46.00	50	49.00
51.00	49.70	46.80	52	50.00
52.00	51.10	47.50	51	50.00
52.00	51.70	48.30	51	50.00
53.00	52.60	49.30	53	52.50
54.00	53.30	50.30	53	54.00
55.00	52.30	51.10	55	55.00
56.00	52.10	52.20	57	55.00
57.00	53.30	52.80	59	56.00
58.00	55.00	53.70	57	57.00
58.00	56.20	54.40	60	57.50

59.00	57.50	55.30	60	59.00
60.00	58.90	56.40	59	60.00
61.00	60.00	57.30	61	61.00
62.00	61.20	58.20	62	62.50
63.00	62.20	59.50	62	63.00
64.00	63.20	59.80	63	64.00
64.00	64.20	60.80	64	65.00
65.00	65.30	61.80	66	65.00
66.00	66.10	62.60	66	66.00
67.00	66.80	63.40	66	67.50
67.50	68.60	64.70	68	68.00
68.00	69.40	65.30	70	68.00
69.00	70.20	66.20	71	69.00
70.00	71.30	67.60	72	70.00
71.00	72.20	68.40	74	71.00
72.00	73.40	69.30	75	72.50
73.00	74.60	70.80	76	73.00
74.00	75.40	71.80	77	73.50
74.00	76.30	72.20	79	75.00
74.00	72.00	72.60	79	75.00
75.00	78.20	73.70	80	76.50
76.50	78.90	74.40	80	77.50
77.00	80.00	75.30	82	78.50
78.00	80.60	76.00	83	80.00
78.00	81.00	76.90	83	80.00
79.00	81.70	78.00	85	81.00
80.00	82.50	78.70	86	82.00
80.00	83.20	79.60	86	82.50
81.00	84.10	80.00	87	83.00
83.00	84.30	80.20	86	84.00
83.00	84.80	81.20	86	85.00
84.00	85.60	82.10	87	86.00
85.00	86.10	82.40	88	86.50
85.00	87.10	83.20	88	87.50
85.00	88.00	84.20	91	88.00
86.00	88.60	84.80	91	89.00
87.00	89.70	85.90	93	90.00
88.00	90.60	87.10	93	90.00

*Note: Data in gray for Bi-Metal was omitted in the graph

Sample Calculations and Results:

Figure 1.2: Platinum Thermometer Temperature Graph

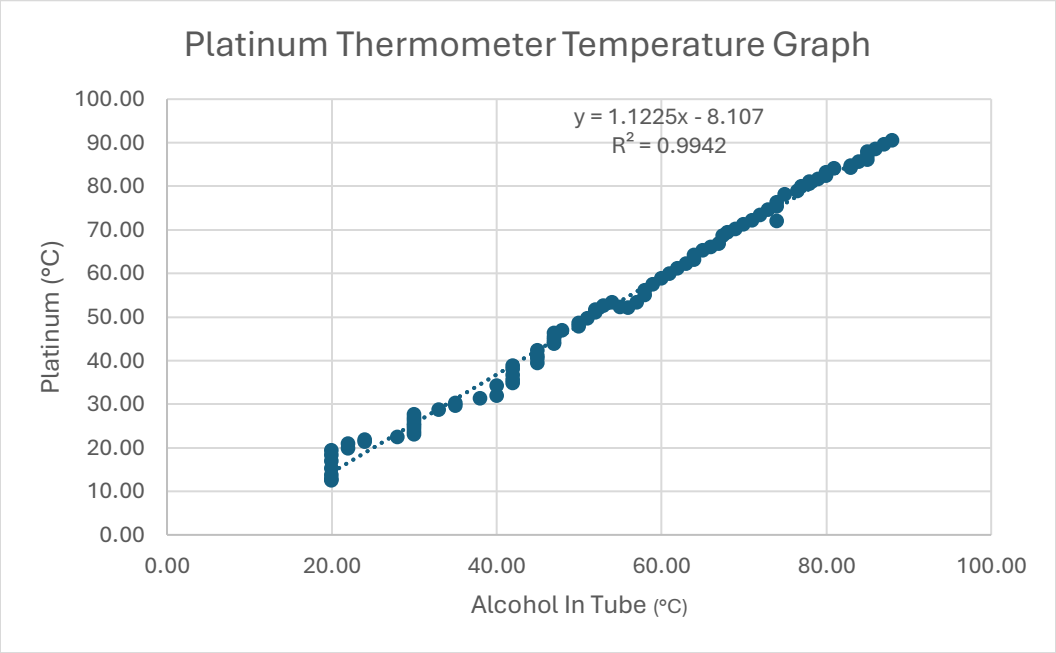


Figure 1.2: Thermistor Temperature Graph

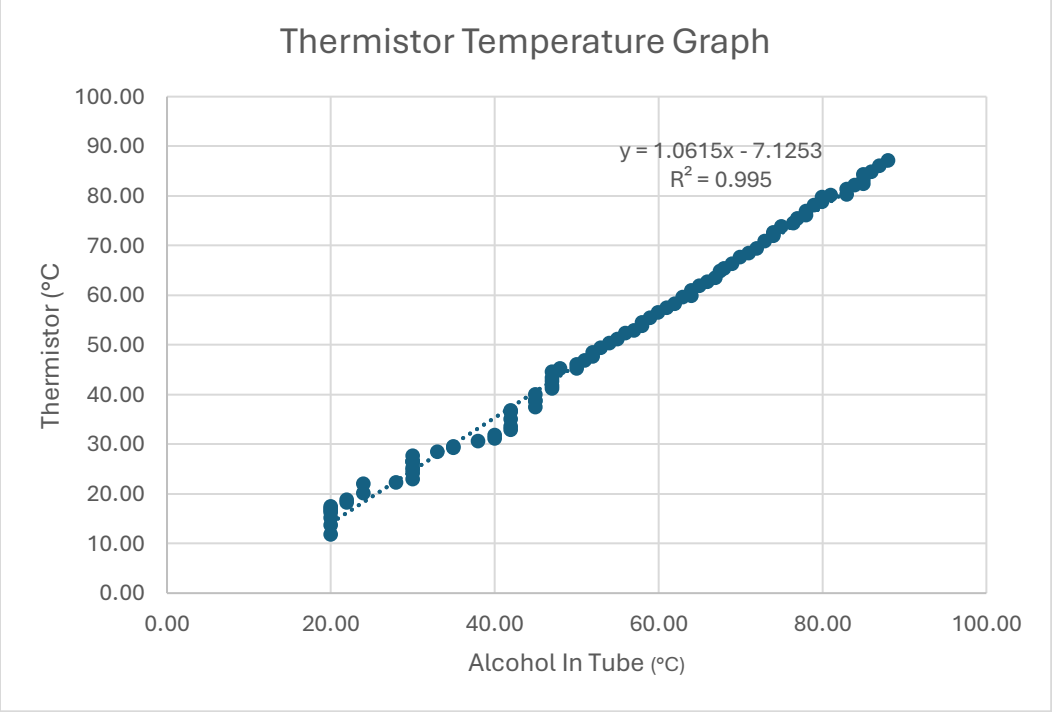


Figure 1.3: Thermocouple Temperature Graph

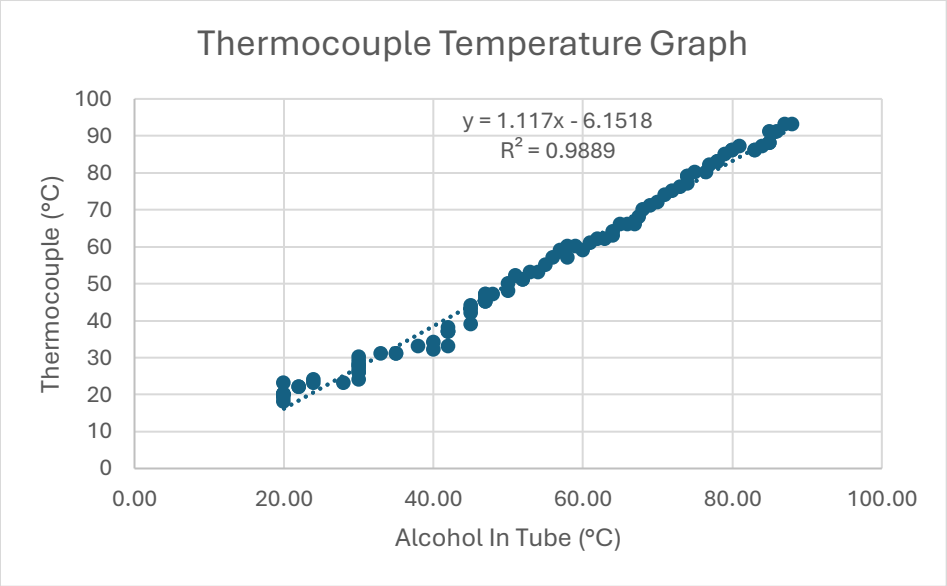


Figure 1.4: Bimetal Temperature Graph

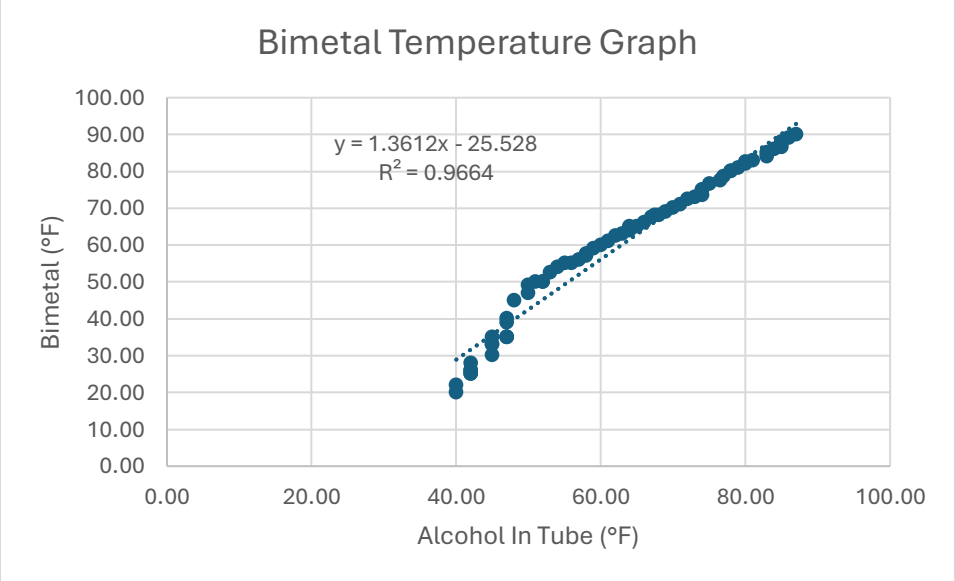


Table 1.1: Thermometer Calibration Calculation Value

	a_0	a_1	r	SE	$y = a_1x + a_0$
Platinum RTD	-8.107	1.1225	0.9942	1.778	$y = 1.1225 - 8.107 \pm 0.5$
Thermistor	-7.1253	1.0615	0.9950	1.5709	$y = 1.0615 - 7.125 \pm 0.5$
Thermocouple	-6.1518	1.1170	0.9889	2.4592	$y = 1.117 - 6.1518 \pm 0.5$
Bi-Metal	-13.368	1.177	0.9664	5.3464	$y = 1.361 - 25.528 \pm 0.5$

Sample Calculations for Platinum RTD:

$$r = \frac{n(\Sigma x_i y_i)-(\Sigma x_i)(\Sigma y_i)}{\sqrt{(n\Sigma x_i{}^2)-(\Sigma x_i)^2)(n\Sigma y_i{}^2)-(\Sigma y_i)^2)}} = \frac{91(294394.5)-(4854)(4710.9)}{\sqrt{(91*297322.5)-4854^2)(91*292549.2)-(4710.9)^2)}} = 0.997$$

$$a_0 = \frac{\Sigma x_i \Sigma x_i y_i - \Sigma y_i \Sigma x_i^2}{(\Sigma x_i)^2 - n(\Sigma x_i)^2} = \frac{4854 * 294394.5 - 4710.9 * 297322.5}{(4854)^2 - 91 * (4854)^2} = -8.107$$

$$a_1 = \frac{\Sigma x_i \Sigma y_i - n\Sigma x_i y_i}{(\Sigma x_i)^2 - n\Sigma x_i^2} = \frac{4854 * 4710.9 - 91 * 294394.5}{(4854)^2 - 91 * 297322.5} = 1.123$$

$$y = a_0 + a_i x = -8.107 + 1.123\text{x} \pm 0.05°C$$

$$SE = \sqrt{\frac{\Sigma(y_i - (a_0 + a_1 x))^2}{n-2}} = \sqrt{\frac{281.4}{91-2}} = 1.778$$

Temperature Calibration for Platinum RTD at time t = 0:

$$x = \frac{(y-a_0)}{a_1} = \frac{(12.5+8.107)}{1.123} = 18.36°C$$

Thermal Response for Platinum RTD:

$$T_{63.2\%\,SC} = T_i + 0.632 * (T_f - T_i) = 18.36 + 0.632 * (87.93 - 18.36) = 62.33\ °C$$

$$t_c = 224 + \frac{62.33-61.74}{62.63-61.74}(228 - 224) = 226.64\ seconds$$

Figure 1.5: Platinum RTD Corrected Temperature Graph

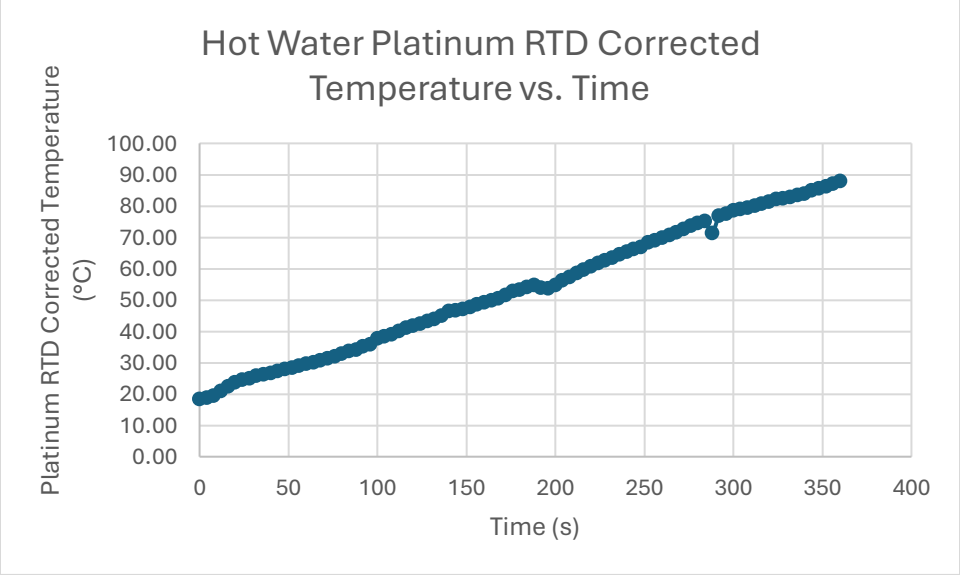


Figure 1.6: Thermistor Corrected Temperature Graph

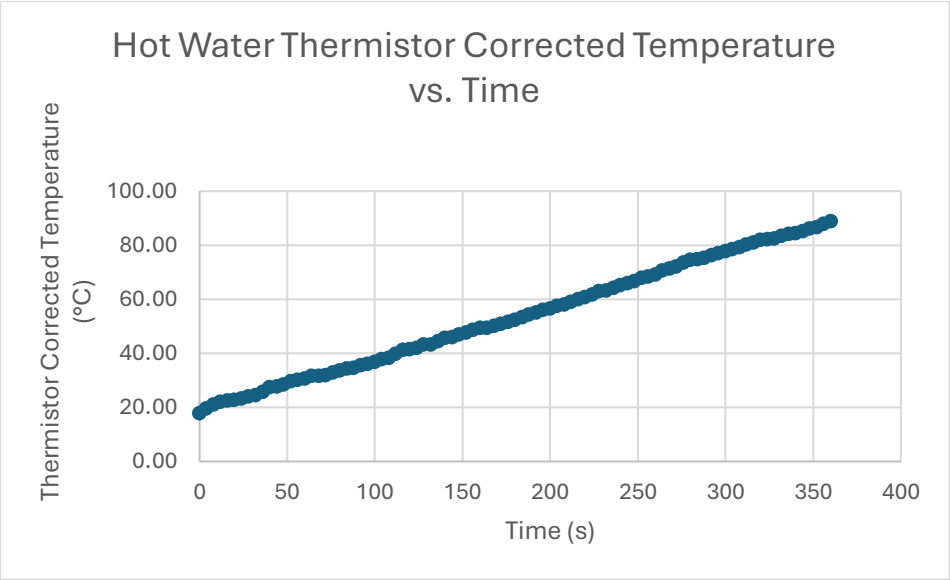


Figure 1.7: Thermocouple Corrected Temperature Graph

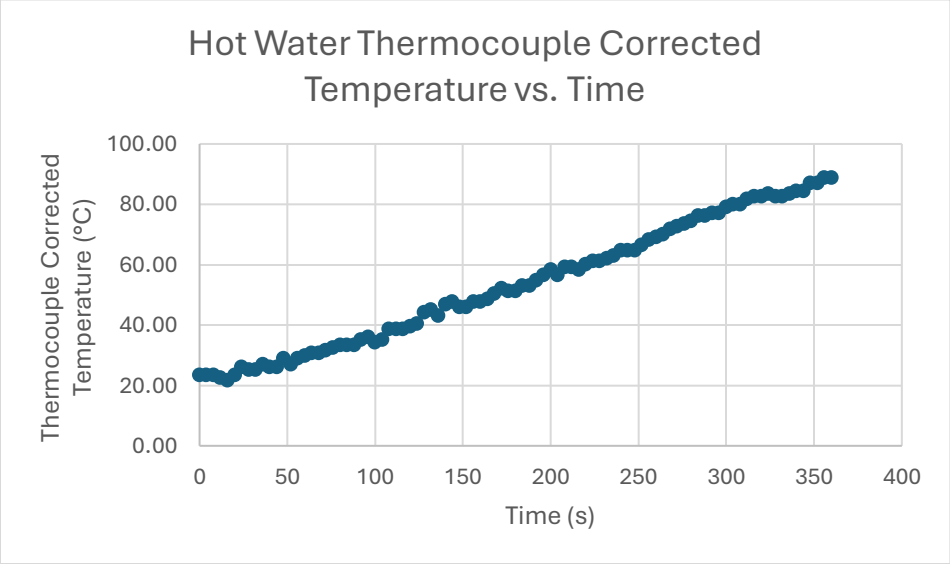


Figure 1.8: Bi-Metal Corrected Temperature Graph

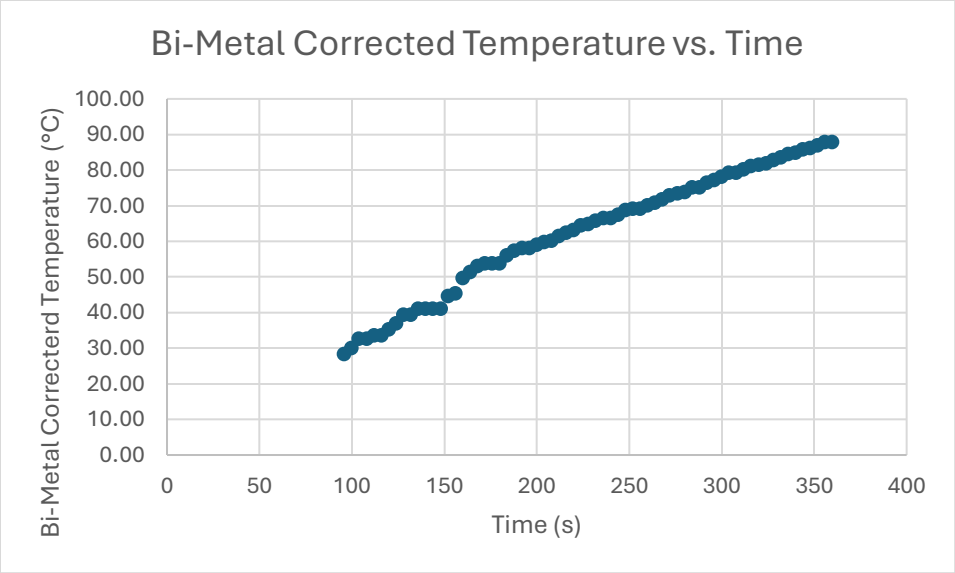


Table 1.2: Time Constant and Step-Change Temperature

Thermometer	$T_{63.2\%\,SC}$ **(°C)**	t_c **(s)**
Platinum RTD	62.330	226.637
Thermistor	59.353	214.109
Thermocouple	66.136	251.441
Bi-Metal	64.24	223.425

Discussion of results:

Part I: Calibration Equations:

For each of the five different types of thermometers used within the experiment, the temperature of each of the devices in the boiling water were recorded. A minimum of 90 data points were required for each of the different thermometers and the method of least squares was utilized to calculate the calibration equations (1), correlation coefficients (6), and the standard errors (5) for the platinum RTD, thermistor, thermocouple, and bi-metal thermometers with respect to the alcohol-in-glass thermometer. For the platinum RTD, the calibration equation is $y = 1.1225 - 8.107 \pm 0.5$, with a correlation coefficient (r) of 0.9942, and standard error of 1.778. For the thermistor, the calibration equation is $y = 1.0615 - 7.125 \pm 0.5$, with $r = 0.9950$, and a standard error of 1.5709. For the thermocouple, the calibration equation is $y = 1.117 - 6.1518 \pm 0.5$, $r = 0.9889$, and a standard error of 10.87506. The bi-metal thermometer had a calibration equation of $y = 1.177 - 13.368 \pm 0.5$, $r = 0.9530$, and a standard error of 5.3464.

Comparing the a_0 values for each of the four different thermometers, each of the values range from -13.368 to -6.1518. Each of the values for a_0 are negative which indicates that each

of the calibration equation lines have a negative intercept. This means that there was a zero-shift error, which means that the output values are inconsistent with the input value from the alcohol-in-glass thermometer. The bi-metal thermometer had the largest negative value with $a_0 = -13.368$, while the thermocouple had the least negative value with $a_0 = -6.1518$. This indicates that the bi-metal had the largest zero-shift error from the alcohol-in-glass thermometers' initial values. However, each of the thermometers still had relatively large negative values for a_0. A possible source of this systematic error may be related to the experimental procedure. Two groups of students used the same measuring instruments consecutively with minimal time between trials. The instruments were therefore subjected to rapid temperature changes, from boiling water to ice cooling and then back to boiling conditions. These large and repeated temperature fluctuations may have prevented the instruments from fully stabilizing, potentially contributing to the observed negative shifts in the calibration equation intercepts for the four thermometers. The values of a_1 had a smaller variation between them as they ranged from 1.0615 to 1.177.

The correlation coefficients were all positive and relatively close to a value of 1. Because each of the correlation coefficients were positive, it means that the direction of their values are the same as the alcohol-in-glass values. This indicates that each of the graphs had a strong linear relationship to the alcohol-in-glass values. The thermistor had $r = 0.9950$, which indicates that it had the strongest linear relationship to the alcohol-in-glass values. It also had a smaller negative value for a_1, which meant that it had less of a zero-shift error than the other graphs. The same can be said for the thermocouple as $r = 0.9889$ and it had the least negative value of a_1. The two graphs with the smaller correlation coefficient were bi-metal and platinum RTD. Observing each of their graphs show that their values differ from the calibration equation line significantly more than the thermistor and thermocouple. The bi-metal plot has a large gap between its values at the beginning half of the graph, with its later values aligning with the calibration equation line. The large gap was caused by the instrument not providing measurement readings until the alcohol-in-glass measurements read 40 °C. The graph for the platinum RTD had slight variations above and below the calibration equation line throughout the whole graph, with its values in the middle being closer to the calibration line.

The bi-metal had the largest standard error ($SE = 5.3464$) compared to the other thermometers. This indicates that its values had the largest variation from the alcohol-in-glass input values. This high standard error can be seen in its graph as there is a large plateau in the data at the beginning of the plot. The bi-metal's high standard error can be attributed to the composition of the instrument as it is made of two different types of metals with different coefficients of linear expansion. Because the measurements on the instrument are reliant on a physical deformation of the internal metal strips, repeated use and improper storage can result in inaccurate measurements. The thermistor had the smallest standard error ($SE = 1.5709$) which means that its values had the smallest variation from the alcohol-in-glass measurements.

Part II & III: Time Constants:

The second and third part of this experiment's calculation found the thermal response time for the platinum RTD, thermistor, thermocouple, and bi-metal thermometers. To find the thermal response time, the conversion of the temperature measurements for each of the four thermometers into equivalent alcohol-in-glass temperatures (7) was required. These "corrected temperatures" were used to plot the corrected temperature versus time graphs. Using equation (8) the temperature that corresponds to 63.2% of a step change in the corrected temperature were calculated. Linear interpolation (9) between the corrected temperatures would then be used to find the time constant. This time constant represents the time it takes a system to change by 63.2% of the total difference between the initial and final temperatures. A smaller time constant means that it takes a system less time to reach 63.2% of its final temperature which indicates a faster response time.

The thermometer with the smallest time constant ($t_c = 214.109\ s$)was the thermistor while the thermocouple had the largest time constant ($t_c = 251.441\ s$). That means that the thermistor was the first device to reach the final steady-state temperature of 90 °C. Having a smaller time constant means that the thermistor had the fastest response time and was the fastest in reading the temperature as it changed. The thermistor has a very fast and non-linear response to thermal change which explains why it had the smallest time constant. The thermocouple had the largest time constant which means that it took the longest to reach 90 °C. The platinum RTD had the second largest time constant ($t_c = 226.637\ s$) which can be attributed to the fact that it is protected with metal shell, which would inhibit its thermal response. The bi-metal had a similar value for its time constant ($t_c = 223.425\ s$) which could be caused by the fact that its measurements come from the deformation of two varying metal strips.

Conclusions and Recommendations:

In conclusion, statistical analysis is an invaluable process to help engineers determine the best types of measuring instruments for certain projects. Within this experiment, five different types of thermometers were used to record the change in temperature of boiling cold water. The resulting calibration equations, correlation coefficients, and standard errors provided valuable insight into the accuracy and errors within each thermometer. This type of analysis using the method of least squares is helpful in determining the variation of a certain measuring instruments from the values of another instrument. In addition, the calculation of the time constants for four of the thermometers, using the alcohol-in glass as the reference allowed for the comparison of the thermometers. Time constants are important when working with apparatuses that measure temperature, as the time constant provides information on the response time of a certain instrument. In comparing the values of the time constants for each of the different thermometers, the thermistor was identified as the thermometer with the fastest response time. Conducting various types of calculations for different instruments is an invaluable process that provides insight into the limitations of certain instruments in certain environments or experiments.

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